

Grade 5 Comprehensive Mathematics Mastery & Foundation Exam (Set 9)

Name: _____

Class: _____

Student ID: _____

Time: 45-50
MinsTotal: 100
Pts

Score: _____

Part 1: Conceptual Precision & Number Sense (Fill-in-the-Blanks & Comparison)**5 Questions, 20 Points****1. Quotient Invariance & Decimal Point Scaling (4 pts)**Given $48 \div 16 = 3$, write down the quotients directly using the rules of division:

(1) $4.8 \div 16 =$ _____ (2) $0.48 \div 16 =$ _____ (3) $0.048 \div 16 =$ _____

(4) $480 \div 16 =$ _____

2. Comparing Operations with Decimals Less Than 1 (4 pts)Fill in each $\square$ with ">", "<", or "=" without detailed calculation:

(1) $6.4 \times 0.8 \square 6.4 \div 0.8$ (2) $12.5 \div 0.5 \square 12.5 \times 0.5$ (3) $0.95 \times 1.02 \square 0.95 \div 1.02$

(4) $8.4 \div 1 \square 8.4 \times 1$

3. Recurring Decimals & Remainder Cycles (4 pts)

(1) Compare: $0.714714 \square 0.\dot{7}1\dot{4}$.

(2) The fraction $\frac{1}{7} = 0.\dot{1}4285\dot{7}$ has a repeating period of _____ digits. The 100th digit after the decimal point is _____.**4. Generality of Algebraic Variables (4 pts)**A square has side length a cm:(1) Its perimeter is _____ cm; (2) Its area is _____ cm^2 ; (3) When $a = 6$, its area is _____ cm^2 .**5. Multiplicative Comparison Sentence Structures (4 pts)**

(1) "How many times 15 is 45?" Equation: _____, Result: _____

(2) "A car traveled 240 km, which is 3 times the distance a bike traveled." The bike's distance is _____ km.

Part 2: Algebraic Simplification & Sign Rules (True/False & Evaluation)**3 Questions, 15 Points****6. Combining Like Terms vs. Constants (5 pts)**True or False: The algebraic expression $4x + 7 + 2x$ simplifies to $13x$. (_____)**7. Additive Inverses & Sign Properties (5 pts)**True or False: The identity $x + x - y + y = 2x - 2y$ is always correct. (_____)

8. "Simplify First, Then Substitute" Discipline (5 pts)

Given $y = 3.5$, evaluate the algebraic expression $15y - 7y + 20$.

Part 3: Rigorous Column Arithmetic & Mixed Operations**3 Questions, 20 Points****9. Column Division with Decimal Divisors & Quotient Alignment (6 pts)**Calculate: $197.6 \div 5.2$

(1) **Pre-calculation estimate:** Round 197.6 to _____, 5.2 to _____, estimated quotient $\approx$ _____.

(2) **Column division :**

$$5.2 \overline{) 197.6}$$

Verification Check:

Is quotient an integer or decimal?

Final result =

10. Triangle Area Formula & Rigor (6 pts)A right triangle has legs of base $b = 0.36$ m and height $h = 0.25$ m.

(1) Write the triangle area formula: Area = _____

(2) Calculate the area of this triangle (show step-by-step working and units):

Answer:

11. Order of Operations with Decimals (8 pts)

(1) $0.4 \times (3.2 - 0.8) \div 1.2$

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(2) $7.4 \times 1.3 - 4.68 \div 0.9$

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=

Part 4: Word Problems & Keyword Modeling**4 Questions, 25 Points****12. Two-Step Equal Sharing & Deducting Remainder (6 pts)**

A bakery made a batch of cookies weighing 9.2 kg in total. The baker distributed them equally into 4 gift boxes, leaving 0.4 kg of cookies on the counter. How many kilograms of cookies were placed in each gift box?

Guided thinking: Total cookies actually packed into the boxes = _____ kg.

Equation & Working:

Answer:

13. Multi-Variable Matching & Unit Alignment (6 pts)

At a supermarket, apples cost 6.8 dollars per kg and bananas cost 4.5 dollars per kg. Ms. Wang bought 15 kg of apples and 24 kg of bananas for the school cafeteria. How much did she pay in total?

Equation & Working:

Answer:

14. Average Score Total Delta Model (6 pts)

The school girls' basketball team averaged 42 points in their first 6 tournament games. They then played 2 more games, raising their overall average across all 8 games to 45 points.

(1) How many total points did the team score in the last 2 games combined? (2) What was their average score in the last 2 games?

Step-by-step Working:

Answer: (1) Total points in last 2 games: _____ pts; (2) Average in last 2 games: _____ pts.

15. Sharing Pitfall & Fraction of the Whole (7 pts)

"Emma bought a large pizza for lunch and cut it into 8 equal slices. She wants to share the pizza evenly between **herself** and **three** of her best friends."

(1) How many people are sharing the pizza altogether? _____ people.

(2) What **fraction** of the whole pizza will each person receive? _____ (fraction) = _____ (decimal).

(3) How many **slices** of pizza will each person get? _____ slices.

16. Bar Model Total Length Conservation Equation (5 pts)

The bar models below show three ribbons of identical total length: Ribbon 1 is $3x$, Ribbon 2 is $x + 8$, and Ribbon 3 is $4y$.

[Tier 1] |---- x ----|---- x ----|---- x ----|
 [Tier 2] |---- x ----|----- 8 -----|
 [Tier 3] |-- y --|-- y --|-- y --|-- y --|

(1) Since Ribbon 1 and Ribbon 2 have equal length, set up an equation in terms of x and solve:

Equation: _____ $\Rightarrow x =$ _____

(2) Find the total length of each ribbon, set up an equation for y , and solve:

Total length: _____ dm; Equation for y : _____ $\Rightarrow y =$ _____

17. Road Paving Collaboration Problem (5 pts)

Engineering teams A and B work together to pave a highway measuring 125 km in total. Team A paves 3.5 km per day, and Team B paves 4.5 km per day. Team A worked alone for the first 6 days due to scheduling, after which Team B joined to work together. How many days will it take to finish the remaining highway after Team B joins?

Answer:

18. Traveling Towards Each Other Equation (5 pts)

City A and City B are 540 km apart. A coach and a truck depart simultaneously from City A and City B respectively, traveling towards each other. They meet after 4.5 hours. The average speed of the coach is 64 km/h. Let the truck's speed be x km/h. **Set up an algebraic equation and solve for x .**

19. Surplus and Shortage Algebraic Model (5 pts)

School organized Grade 5 teachers and students for a boating activity in a park. If each boat holds 4 people, 6 people are left without a boat. If each boat takes 2 more people (i.e. 6 people per boat), exactly 1 boat is completely empty. How many boats were originally rented? How many total teachers and students participated?



[Answer Key & Marking Rubric] (Teacher Reference)

Part 1 (20 pts): 1. (1) 0.3 (2) 0.03 (3) 0.003 (4) 30 (1 pt each); 2. (1) $<$ (2) $>$ (3) $<$ (4) $=$ (1 pt each); 3. (1) $<$ (2 pts) (2) 6, 8 (1 pt each); 4. (1) $4a$ (2) a^2 (3) 36 (1+1+2 pts); 5. (1) $45 \div 15$, 3 (2 pts) (2) 80 (2 pts).

Part 2 (15 pts): 6. False (2 pts); Simplification: $6x + 7$ (2 pts); Reason: Only like terms can combine coefficients, constant 7 cannot merge with x (1 pt). 7. False (2 pts); Steps: $(x + x) + (-y + y) = 2x + 0 = 2x$ (3 pts). 8. Simplify: $8y + 20$ (2 pts); Evaluate: $8 \times 3.5 + 20 = 28 + 20 = 48$ (3 pts).

Part 3 (20 pts): 9. (1) Estimate: $200 \div 5 = 40$ (2 pts); (2) Column division: quotient is integer 38 (4 pts; deduct 3 pts if misplaced decimal point). 10. (1) $S = \frac{1}{2}bh$ (2 pts); (2) $0.36 \times 0.25 \div 2 = 0.045 \text{ m}^2$ (4 pts; deduct 3 pts if forgot to divide by 2). 11. (1) $0.4 \times 2.4 \div 1.2 = 0.96 \div 1.2 = 0.8$ (4 pts); (2) $9.62 - 5.2 = 4.42$ (4 pts).

Part 4 (25 pts): 12. Guide: 8.8 kg (1 pt); $(9.2 - 0.4) \div 4 = 2.2 \text{ kg}$ (4 pts); Statement (1 pt). 13. $6.8 \times 15 + 4.5 \times 24 = 102 + 108 = 210$ dollars (6 pts). 14. (1) $45 \times 8 - 42 \times 6 = 360 - 252 = 108$ pts (4 pts); (2) $108 \div 2 = 54$ pts (2 pts). 15. (1) 4 people (2 pts); (2) $1/4$, 0.25 (2 pts); (3) 2 slices (3 pts).

Part 5 (20 pts): 16. (1) $3x = x + 8 \implies 2x = 8 \implies x = 4$ (2 pts); (2) Total length 12 dm (1 pt), $4y = 12 \implies y = 3$ (2 pts). 17. Remaining: $125 - 3.5 \times 6 = 104 \text{ km}$; $104 \div (3.5 + 4.5) = 13$ days (5 pts). 18. Equation: $(64 + x) \times 4.5 = 540 \implies 64 + x = 120 \implies x = 56 \text{ km/h}$ (5 pts). 19. Expressions: $4x + 6$ and $6(x - 1)$ (2 pts); Equation: $4x + 6 = 6(x - 1) \implies 2x = 12 \implies x = 6$ boats (2 pts); Total people: $4 \times 6 + 6 = 30$ people (1 pt).